

Attributes, Abstractions, and Combinations

A New Taxonomy for Studying How Parties Invoke Social Groups

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Supplementary Material

Supplemental materials – including additional details on the dataset, mention detection and attribute classification procedures, and additional figures, tables, and robustness checks – are included as an appendix to this document.

Data Availability Statement

The data and code supporting the findings of this study are openly available in Harvard Dataverse at DOI TO BE ADDED.

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Abstract

Research increasingly examines which social groups political actors refer to. Yet, existing approaches typically inherit classifications shaped by demand-side perspectives or conflate how groups are defined with how they are framed. We propose a novel, attribute-centered taxonomy of social group references that classifies groups by the attributes political actors invoke, organizes these attributes along nested levels of abstraction, and captures attribute combinations when multiple attributes are invoked within a single reference. To demonstrate the analytical value of our taxonomy, we collect original annotations and develop automated classification methods to quantify the group focus strategies of parties in 36 countries over more than 50 years. Our application underscores that socio-structural, constructivist, and deservingness perspectives on party competition primarily operate at different – though overlapping – analytical levels, and yields substantive insights into how different party families construct and mobilize social groups. Our taxonomy, data, and pre-trained classifiers contribute new resources to facilitate future research on social group references in political discourse.

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1 Introduction

Social groups are fundamental building blocks of democratic representation. Party–group alignments have historically been stabilized not only by shared interests but also by collective identities that connect citizens to political parties. During the era of classical cleavage politics, these alignments were formed and sustained through dense organizational networks and socialization within relatively stable social milieus (Lipset and Rokkan 1967; Bartolini and Mair 2007). While these organizational foundations have substantially weakened, social groups and group-based political identities have not disappeared. Instead, contemporary processes of partisan realignment demonstrate the continued formation of durable political alignments around social groups and identities (Bornschiefer 2010; Hooghe and Marks 2018; Gethin et al. 2021; Westheuser and Zollinger 2025). In this less organizationally embedded environment, political communication becomes a particularly important arena through which parties invoke and frame social groups, articulate shared interests and identities, and seek to forge and sustain ties with social groups. Understanding how parties communicate about social groups is therefore pivotal for understanding political representation under conditions of partisan realignment.

Yet measuring how political parties invoke groups presents a surprisingly difficult problem. Political texts contain an enormous variety of group references: UK Labour and Conservative manifestos alone contain more than 2,700 unique social group mentions between 1964 and 2015 (Thau 2019), ranging from broadly defined collectives, such as “the people,” to highly specific groups, such as “Nobel Prize winners.” Recent computational approaches increasingly allow researchers to detect such references at scale. But detection alone does not make these heterogeneous expressions comparable.

This leaves an important measurement gap on the supply side of political representation. Despite a rapidly growing literature on party references to social groups (e.g. Thau 2019; Dolinsky and Huber 2023; Dausgaard and Hjorth 2024; Riethmüller and Franzmann 2025; Amarales et al. 2025; Horne et al. 2025; Licht and Sczepanski 2025; Dolinsky et al. 2025), existing approaches face a trade-off between comparability and openness to supply-side group construction. Approaches based on predefined categories provide a common structure for

comparison but often draw on particular theories or demand-side classifications, potentially imposing assumptions about which groups matter and overlooking how parties define, combine, and construct social groups in political communication. More inductive approaches avoid such assumptions by recovering group references directly from political texts, but leave researchers with a highly heterogeneous set of specific groups that is difficult to compare systematically across parties, political contexts, and time. What is missing is a common framework that combines both advantages: remaining open to the groups parties themselves invoke and construct while organizing these references in a form suitable for systematic comparative research.

We address this gap by developing a taxonomy of social group references. Building on the insight that political actors invoke social groups through attributes rather than fixed group categories (Licht and Szczepanski 2025; Chandra 2012), we make attributes the organizing principle of a general taxonomy for studying parties' group-based communication. Rather than beginning with predefined social groups, the framework decomposes group references into the attributes through which parties invoke and define them in political communication. It captures how parties construct politically meaningful social groups by combining multiple attributes, organizing them across nested levels of abstraction. Together, these features provide a common measurement framework for comparative research on party communication while leaving questions of identity, deservingness, redistribution, or other forms of political evaluation analytically distinct from the classification of social group references. The resulting taxonomy provides a common analytical language that supports cumulative research while remaining compatible with competing theories of political representation, party competition, and group formation.

Our taxonomy rests on three principles. First, it represents group references through the attributes political actors explicitly invoke rather than through predefined group or identity categories. This allows researchers to identify both expected and novel forms of group construction without assuming *ex ante* which social groups are politically relevant. Second, it organizes attributes across levels of abstraction, allowing researchers to move systematically between broad dimensions of social differentiation and increasingly specific group references while preserving comparability across studies. Third, it recognizes that

parties frequently define groups through combinations of attributes. Rather than forcing mutually exclusive classifications, the taxonomy therefore captures attribute composition by assigning all defining attributes to a single group reference.

We operationalize our taxonomy through a scalable two-step measurement procedure. Building on recent advances in automated group mention detection (Licht and Sczepanski 2025), we first identify verbatim social group references in party manifestos. We then classify the group-defining attributes contained in these references using novel supervised classifiers developed for the taxonomy introduced in this paper.¹ Together, the resulting classifiers enable the comparative analysis of party group references across parties, countries, and time at a scale that would be infeasible through manual coding alone.

We illustrate the analytical leverage of our framework using a corpus that combines broad longitudinal coverage of Green and populist radical right (PRR) parties across 36 democracies with Social Democratic and Conservative manifestos from Germany, Sweden, the United Kingdom, and the United States. This design deliberately emphasizes parties that have played a central role in disrupting established party-group alignments, while using established party families to provide comparative benchmarks. Supplementary analyses using the Manifesto Corpus Translations corpus (Ivanusch and Regel 2024) show that the group-reference profiles of the selected Social Democratic and Conservative parties closely resemble those of their broader party families. The resulting classifications reveal which social groups parties invoke across levels of abstraction and attribute categories. They show that group references frequently combine multiple attributes, that important differences between party families emerge at increasingly specific levels of abstraction, and that these patterns of group construction evolve alongside broader processes of partisan realignment. Focusing on Green and PRR parties, we further demonstrate how the taxonomy can address questions derived from socio-structural and constructivist theories directly, while also providing the measurement foundation for complementary analyses of group evaluation and deservingness. By making the taxonomy, annotated data, and measurement pipeline transparent and available, we also enable future research to extend the framework to additional parties and party families.

¹The models are publically available for download at huggingface.co/collections/haukelicht/social-group-mention-attribute-classification.

Our paper makes three contributions. First, we develop a general, attribute-centered taxonomy of social group references that provides a common framework for studying group-based political communication comparatively while remaining agnostic with respect to competing theories. Second, we introduce a scalable measurement framework that operationalizes this taxonomy through automated group mention detection and attribute classification. Third, we demonstrate the analytical leverage of this framework by showing how our taxonomy can inform socio-structural, constructivist, and deservingness perspectives on party competition.

2 Categorizing social group references

Studying social group references in political communication requires a framework for organizing a highly diverse set of linguistic expressions. Party manifestos refer to groups ranging from broad collectives such as *society*, *citizens*, or *the people* to highly specific references such as *Nobel Prize winners*, *small business owners*, or *young unemployed workers*. Even within a single party system, the diversity of these references is substantial. For example, UK Labour and Conservative manifestos between 1964 and 2015 contain more than 2,700 unique social group mentions (Thau 2019). The central challenge in studying group references in political communication is therefore not merely to detect group references (cf. Licht and Sczepanski 2025), but to organize them into a coherent analytical framework that supports systematic comparison across parties, countries, and time.

Recent research has made considerable progress in measuring parties’ group appeals. Existing approaches classify group references according to representational claims (De Mulder et al. 2025), positive or negative evaluations (Riethmüller and Franzmann 2025; Horne et al. 2025), identity and non-identity groups (Stuckelberger and Tresch 2024), social classes (Thau 2021), or specific socio-demographic categories (Huber 2021; Huber et al. 2024; Dolinsky and Huber 2023; Dolinsky et al. 2025). These approaches have generated important insights for their respective research questions. At the same time, they differ substantially in how they define and organize social group references, making comparisons across studies increasingly difficult as the literature expands.

Despite the differences among the various approaches to categorizing social group men-

tions in political texts, three recurring conceptual challenges emerge. The first challenge concerns the **fundamental unit of classification**. Most existing approaches begin with predefined social groups such as *workers*, *immigrants*, *women*, *pensioners*, or *ethnic minorities*. These classifications are typically derived from survey research, structural theories of social stratification, or particular substantive research traditions (e.g. Dolinsky and Huber 2023; Huber 2021; Riethmüller and Franzmann 2025). While well suited to specific theoretical questions, they inevitably embed assumptions about which social groups are politically relevant. From the perspective of political communication, however, parties do not merely invoke pre-existing social groups; they may redefine, combine, or construct social groups in ways that extend beyond existing classifications. This raises the question of whether a more fundamental unit of classification is required that allows both familiar and emerging social groups to be represented systematically.

The second challenge concerns the relationship between **group definitions and their political framing**. Existing classifications frequently intertwine the attributes used to define a group with the political meaning attached to it. However, attributes answer the question “Who is the referenced group?”, whereas frames answer “How is that group interpreted and portrayed?” (cf. Entman 1993). This analytical distinction does not imply that group definition itself is outside the broader process of framing: parties may already frame political conflict by selectively invoking some attributes and groups rather than others.

This distinction becomes particularly important when classifications infer how groups are politically interpreted from the attributes through which they are defined. Occupational groups, for example, are frequently treated as inherently economic constituencies, whereas groups defined by gender, ethnicity, or religion are commonly interpreted as identity groups. Yet these associations are not conceptually necessary and may even hinder comparability across studies. The same social group may be framed through material interests, collective identities, deservingness, moral values, recognition, or political grievances. Class politics, for example, depends not only on the hierarchical position in the social structure but also on parties’ strategic articulation of class as an identity (Przeworski and Sprague 1986; de Leon 2014; Thau 2021). Likewise, feminist scholarship has long argued that gender injustice simultaneously concerns recognition and redistribution (Fraser 2013). Similarly, Polanyi’s

(1944) account of the “double movement” illustrates that economic conflicts are inseparable from broader social and political orders. Taken together, these perspectives indicate that the attributes through which parties define social groups should be distinguished analytically from how these groups are subsequently interpreted and evaluated. Both processes may constitute political framing, but they represent analytically distinct aspects of party communication.

The third challenge concerns **how social group references should be organized**. Political communication contains references that differ widely in their level of specificity, ranging from broad collectivities such as *society* to highly specific groups such as *working mothers*, *young entrepreneurs*, or even *persons who receive a residence permit as part of family reunification or on a humanitarian basis*. Existing approaches frequently mix categories with varying breadth, making aggregation and comparison across studies difficult. Moreover, political actors often define groups through combinations of multiple attributes rather than a single defining characteristic. A general taxonomy must therefore organize group references across common levels of abstraction while accommodating the complexity of groups constructed through combinations of attributes.

3 From classes and identities to attributes

We address these three challenges through an attribute-centered taxonomy of social group references. Our taxonomy is built on three design principles: (1) it takes attributes rather than predefined social groups as its fundamental unit of categorization; (2) it organizes these attributes across nested levels of abstraction; and (3) it captures attribute combinations by allowing multiple defining attributes to be assigned to a single group reference. Together, these principles provide a common framework for studying the references to and construction of social groups in political communication. The taxonomy deliberately focuses on identifying and organizing group-defining attributes, thereby establishing a common analytical foundation for subsequent analyses focusing on, for example, political framing, communicative coalition building, or elite-driven identity construction.

The first design principle of our taxonomy concerns the fundamental unit of classification. We represent social group references through the attributes political actors invoke rather

than through predefined social groups or identity categories (cf. Licht and Szczepanski 2025).² We define an attribute as a property of persons – or of the relations between persons – that can be used to define membership in a collectivity. A single attribute may define a group, as in *workers* or *women*, while multiple attributes may jointly define a social group, as in *young farmers* or *working mothers*. The defining characteristic of an attribute is therefore not its substantive content but its capacity to delineate group membership through political communication.

Representing groups through attributes has important methodological advantages. Because attributes rather than predefined groups constitute the unit of classification, familiar and novel group constructions can be represented within the same framework. Further, it separates group identification from framing, which is particularly important because class, gender, ethnicity, religion, and other attributes have all been shown to support multiple forms of political interpretation, including redistribution, identity, recognition, deservingness, and moral evaluation (e.g. Przeworski and Sprague 1986; de Leon 2014; Fraser 2013; Zollinger 2022, 2024; Szczepanski 2023). Finally, an attribute-centered perspective accommodates both stable and emerging forms of political group alignment. Socio-structural approaches understand social groups primarily as products of enduring positions within social relations and institutions (Lipset and Rokkan 1967; Dalton 1984, 2008; Kriesi et al. 2008, 2012; Piketty 2018; Gethin et al. 2021). By contrast, constructivist and entrepreneurial accounts emphasize that political actors actively constitute political groups by selecting, combining, and reinterpreting social attributes, while occasionally introducing novel forms of social differentiation (Chandra 2012; Wolkenstein and Wratil 2021; Zuber et al. 2025; Damhuis 2020; Zollinger 2022, 2024; Westheuser 2025; Westheuser and Zollinger 2025). Because our taxonomy takes attributes rather than predefined groups as its point of departure, it remains agnostic as to whether the resulting social groups primarily reflect enduring social structures or political group construction.

²Our emphasis on attributes builds on Chandra’s (2012) distinction between *attributes*, the *identity categories* constructed from them, and broader *category dimensions*. While Chandra develops this distinction to explain the construction of ethnic identities, we adopt attributes as the organizing principle of a general taxonomy for social group references. Our objective is not to explain a particular class of identities but to provide a measurement framework applicable across all forms of party group appeals.

3.1 Levels of attribute abstraction

The second design principle of our taxonomy is that we organize attributes hierarchically according to their level of abstraction. Parties invoke groups ranging from specific (“nurses”, “single mothers”) to highly abstract ones (“people”, “humankind”). Shifting abstraction lets actors broaden or narrow appeals (Damhuis and Karremans 2017; Evans and Tilley 2017; Horn et al. 2021). Classic theories imply this logic. Pitkin (1967) distinguishes concrete constituency representation from abstract claims to general interests. Downs (1957) notes that vote-seeking parties should use inclusive appeals. In realignment periods, parties invoke broad groups that cut across weakened cleavages (Kriesi et al. 2008, 2012; Dalton 2008; Damhuis and Karremans 2017; Evans and Tilley 2017; Horn et al. 2021). Research on populism makes this explicit, stressing that “the people”, “the other” and “the elite” are signifiers used to cater to constituencies with common grievances but diverse demands (Laclau 2005; Mudde and Kaltwasser 2017).

Studies on parties’ group references differ in how they treat abstraction. Some use single-level classifications (Huber 2021; Huber et al. 2024; Riethmüller and Franzmann 2025). Others use limited levels of abstraction (Thau 2019; Stuckelberger and Tresch 2024). Usually, abstraction is a coding device, not an analytical dimension. We instead treat levels of abstraction as crucial for systematizing and understanding parties’ social group focus strategies. We distinguish **three nested levels of abstraction**: attribute dimensions, attribute categories, and specific group references.

Attribute dimensions

At the highest level of abstraction, our taxonomy distinguishes attributes according to two simple questions. First, does the reference rely on boundary-making attributes? Second, if so, do these attributes refer to economic positions and activities or to other social characteristics? Together, these decisions yield three broad **attribute dimensions**: universal, economic, and non-economic.

Universal group references invoke collectives without boundary-making attributes. References such as *humankind*, *the people*, or *society* denote collectives but do not specify

characteristics that distinguish members from non-members. By contrast, all remaining group references invoke at least one attribute that defines membership in a particular collectivity. We therefore distinguish universal group references from all other references.

Among boundary-making attributes, we distinguish between economic and non-economic characteristics. Economic attributes refer to individuals' positions or activities in economic life, whereas non-economic attributes refer to other social characteristics such as nationality, ethnicity, gender, age, religion, family status, health, place, or shared values. We adopt this distinction because it provides a widely used organizing principle in comparative politics and political sociology, including research on party competition (Kriesi et al. 2008, 2012), class politics (Thau 2019, 2021), and distributional conflict (Piketty 2018; Gethin et al. 2021). Organizing attributes along these broad dimensions therefore facilitates comparison with existing research while preserving sufficient flexibility to capture diverse forms of political group construction.

These dimensions serve an analytical rather than theoretical purpose: they organize attributes according to the type of social characteristic used to define a group, not according to assumptions about political meaning, framing, or relative importance. Economic attributes may be invoked through identity, moral, cultural, or deservingness frames, just as groups defined by non-economic attributes may be framed in distributive terms; nor does either dimension presuppose social hierarchy (Stewart 2000, 2016) or moral ordering. Which attributes are more politically consequential, and how they are subsequently framed, remain empirical questions.

Finally, the hierarchical structure of the taxonomy is modular rather than prescriptive. Researchers whose theoretical questions do not require the distinction between economic and non-economic attributes can work directly with the attribute categories and disregard the higher-order dimensions. The categories therefore remain usable independently of how researchers choose to aggregate them at higher levels of abstraction.

Attribute categories

While attribute dimensions can address substantive questions at a high level of abstraction, many questions about political communication require more fine-grained attribute categories.

Table 1: Taxonomy of attribute dimensions and categories. Rows are grouped by attribute dimension.

Dimension	Category	Examples
Universal		the people, humankind, all citizens, society
Economic	<i>education</i>	students, apprentices, graduates
	<i>employment status</i>	employers, employees, self-employed, or unemployed people
	<i>income/wealth/economic status</i>	the poor, high-income earners, homeowners/tenants
	<i>occupation/profession</i>	teachers, farmers, public servants, police officers
Non-economic	<i>age</i>	children, young people, old people, future generations
	<i>crime</i>	offenders/criminals or victims
	<i>ethnicity</i>	people of color or ethnic minorities
	<i>family</i>	fathers, mothers, parents
	<i>gender/sexuality</i>	women, men, LGBTQ+ people
	<i>health</i>	disabled/handicapped people or chronically sick people
	<i>nationality</i>	natives or immigrants
	<i>place/location</i>	rural residents, urban dwellers, people from the global south
	<i>religion</i>	Christians, Muslims, Jews, religious minorities
	<i>shared values/mentalities</i>	people with meritocratic values, environmentalists

We therefore introduce a second level of abstraction consisting of **attribute categories**, which organize specific group-defining attributes into analytically coherent and mutually exclusive classes. Table 1 presents the categories covered by our taxonomy and illustrates how these categories are nested within the broader economic and non-economic attribute dimensions.

Each attribute category captures a distinct type of characteristic through which parties define social groups while remaining sufficiently broad to encompass a wide range of concrete group references. For example, the *employment status* category includes references to employees, the unemployed, pensioners, and the self-employed, whereas the *gender/sexuality* category encompasses women, men, LGBTQI+ persons, and other groups defined through gender or sexuality-related attributes. This intermediate level of abstraction enables researchers to compare broad patterns of party communication without losing the richness of individual

group references.

The categories were selected according to three criteria. First, they should be comprehensive, covering the broad range of attributes parties invoke in political communication. Second, they should be mutually exclusive, ensuring that each defining attribute belongs to one category only. Third, they should be hierarchically coherent, meaning that categories capture general types of attributes rather than specific social groups. Consequently, groups such as health professionals or teachers are not treated as separate categories (cf. Dolinsky et al. 2025) but are classified under the broader category of *occupation/profession*.

Importantly, one of our categories extends existing classification schemes: *shared values/mentalities*, which captures groups defined through reference to common moral dispositions, values, lifestyles or worldviews rather than conventional socio-demographic characteristics. References such as “informed, creative, critical, participatory, demanding and above all free citizens” or “every German who can and wants to exercise their freedom” illustrate that parties frequently constitute social groups through shared values or moral dispositions. Recent work on political identities, symbolic boundaries, and entrepreneurial party competition increasingly argues that contemporary conflicts are organized not only around socio-demographic divisions but also around moral and value-based distinctions that define who belongs to “us” and “them” (Lamont 1992, 2009; Brubaker 2004; Damhuis 2020; Zollinger 2022, 2024; Westheuser 2025; Westheuser and Zollinger 2025; Sczepanski 2023). Yet existing taxonomies rarely capture these constitutive value attributes explicitly, while the literature on moral boundaries and symbolic classification has largely relied on qualitative or small-scale analyses that are difficult to compare systematically across parties, countries, and time. As we demonstrate in the empirical application, distinguishing groups defined through shared values or moral evaluations provides important insights into contemporary party competition.

Specific group references

The third and most specific level of the taxonomy consists of **specific group references**. Unlike attribute dimensions and categories, these references correspond to the concrete linguistic expressions through which parties invoke social groups in political texts. They

are inherently more diverse and context dependent. For example, the attribute category *gender/sexuality* encompasses references to women, men, and LGBTQI+ people, while the category *occupation/profession* includes various references to possibly thousands of different occupational and professional groups ranging from manual workers, over white collar workers, to sociocultural professionals and artistic occupations (Oesch 2006). Moreover, the same specific group may be referenced through various, lexically different formulations depending on sociocultural context, rhetorical style, and communication arena.

Research on ethnic politics and social categorization has long emphasized that many social groups are historically contingent and actively constructed through political discourse rather than fixed social entities (Brubaker 2004; Chandra 2012). Parties contribute to defining group boundaries, meanings, and political relevance (Lamont and Molnár 2002; de Leon 2014). We extend this insight beyond ethnic and identity politics by treating all specific group references as empirical expressions of broader attribute categories rather than as fixed analytical units.

Unlike the higher levels of the taxonomy, however, we believe that no single categorization strategy is appropriate for organizing specific group references. In some domains, refined deductive classification systems already exist. Occupational references, for example, can be linked to standardized taxonomies such as ISCO,³ a taxonomy covering four nested levels of abstraction. Other classifications combine multiple attribute categories to derive more elaborate social typologies. Oesch’s class schema (2006), for instance, builds on combinations of occupational characteristics, employment relations, and educational qualifications to distinguish broader class positions. In other domains – including moral dispositions, place-based identities, lifestyles, or emerging political constituencies – specific group references remain highly contingent and may be better organized through inductive or data-driven approaches. We therefore deliberately leave the organization of specific group references open to domain-specific deductive classifications and inductive approaches, depending on the analytical objectives and the maturity of the respective research field.

By separating specific group references from the broader attribute categories to which they belong, the taxonomy provides a common analytical foundation on which both deductive

³ISCO stands for *International Standard Classification of Occupations*.

and inductive classification strategies can build. Existing domain-specific taxonomies can be integrated by mapping them onto combinations of attribute categories, while emerging or context-specific groups can be studied inductively without sacrificing comparability at higher levels of abstraction.

3.2 Attribute combinations

The third design principle of the taxonomy concerns attribute combinations. Political actors frequently combine groups and attributes in different ways to define the constituencies they seek to mobilize. We therefore view attribute combinations as a general feature of political communication that operates at multiple analytical levels.

At the broadest level, parties compose constituencies *across* multiple group references. Coalition theories have long emphasized that parties assemble electoral support by mobilizing heterogeneous social groups (Riker 1962). Historical accounts of class politics similarly show how labour parties combined class appeals with national, religious, gendered, or moral constituencies to construct durable electoral coalitions (Przeworski and Sprague 1986; Eley 2002; Howe et al. 2022). More recent work on partisan realignment demonstrates that contemporary electoral coalitions increasingly combine economic and cultural constituencies (Kriesi et al. 2008, 2012; Piketty 2018; Gethin et al. 2021), while constructivist approaches emphasize how parties actively link heterogeneous social demands into broader political identities (Mudde and Kaltwasser 2017; Zollinger 2024). Our taxonomy naturally facilitates the study of this first form of composition by providing a common language for identifying and aggregating the different social groups parties invoke across speeches, manifestos, or other political texts.

However, political actors also compose constituencies at a second level: *within* individual group references. Rather than merely mentioning different groups, parties frequently define a single constituency through the combination of multiple attributes. References such as *working mothers*, *young farmers*, or *those living and working in rural communities* derive their meaning precisely from the joint presence of several defining attributes rather than from any single attribute alone.

Our notion of attribute combination draws inspiration from the analytical logic underlying

intersectionality research (Crenshaw 1989; McCall 2005), namely the insight that social groups may be constituted through multiple interacting attributes rather than single characteristics in isolation. We deliberately adopt the broader term attribute composition, however, because our framework captures all instances in which political actors combine multiple defining attributes rather than only those associated with systems of social stratification or disadvantage. We opt for using the broader umbrella concept of attribute composition because multi-attribute expressions arise through two distinct communicative logics. In *conjunctive* references, multiple attributes jointly define a single, narrower constituency. The phrase “young farmers”, for example, refers to people who are simultaneously young *and* farmers, while “working mothers” denotes persons who are both employed and mothers. Here, the combination of attributes constitutes the social group itself. By contrast, *enumerative* references simply list separate groups within a single expression.⁴ References such as “workers and mothers” invoke multiple constituencies without combining the attributes into a single group.

Together with attribute dimensions and attribute categories, attribute composition completes our taxonomy. It enables researchers to study not only which social groups parties invoke, but also how they combine – both across multiple group references and within individual group references themselves. In particular, measuring within-reference attribute composition provides a necessary foundation for studying intersectional group references in political communication while simultaneously encompassing the much broader range of compound group constructions found in everyday political discourse.

4 Measuring attribute dimensions and categories in social group mentions

To demonstrate our approach, we apply automated text analysis methods to, first, extract social group mentions from party manifestos, and second, classify these mentions according to the attribute categories they feature.

⁴Enumerative group references should not be understood as arbitrary lists, however. Political actors typically enumerate groups because they are linked by a common political frame or narrative.

4.1 Case selection and data

Our case selection of manifestos was designed for the development of a general taxonomy and measurement framework for social group mentions. Developing such a framework requires training data that captures the variety in which political actors define social groups across political contexts, languages, ideological traditions, and over time. To maximize this diversity, we adopted two complementary sampling strategies.

First, we aimed for the broadest possible coverage of Green and populist radical right (PRR) parties. These challenger party families occupy a critical place in contemporary debates on political representation, identities, and group appeals because they are widely argued to have disrupted established party–group linkages and transformed the social groups mobilized in party competition (Kriesi et al. 2008, 2012; Bornschier 2015; Westheuser and Zollinger 2025). This makes their communication particularly informative for a framework designed to measure group references in periods in which established party–group alignments cannot be taken for granted. Because Green and PRR parties remain comparatively rare and exhibit considerable variation across countries and over time, we sought to include their communication as comprehensively as possible. We have compiled a corpus of the election manifestos of PRR and Green parties across 36 democracies from 1970 until 2021. Manifestos’ texts were taken from secondary sources⁵ and we filled gaps through original data collection wherever possible.⁶

Second, for centre-left and -right parties, we selected Social Democratic and Conservative parties following a diverse-case sampling strategy. Unlike Green and PRR parties, these party families are numerous, historically well established, and comparatively stable across advanced democracies. Rather than seeking exhaustive coverage, we selected all manifestos of Social Democratic and Conservative parties in Germany, Sweden, the United Kingdom, and the United States. These selected countries are diverse in terms of party system, electoral

⁵The *Manifesto Project Dataset* (Lehmann et al. 2023), *PoliDoc* (Benoit et al. 2009), and *ManifestoVault V1.0* (Dolinsky et al. 2024). In many countries, existing collections of parties’ election manifestos (e.g., Lehmann et al. 2023; Benoit et al. 2009) date back into the 1970s and thus we typically cover the entire history of the new right and new left.

⁶In order to classify parties as being PRRPs, we follow the definition of Mudde (2007) and see nationalism as their core ideological feature. For Green parties we follow Röth and Schwander (2021) or Dobson (2012) who define Green parties by a strong emphasis of ecologism and environmentalism and typically, but not always, incorporate the term “Green” in their party names.

institutions, and political context. This strategy increases the diversity of the training data while remaining feasible given the substantial manual effort required for iterative taxonomy development, multiple rounds of multi-label annotation, annotation revision, and classifier validation. However, using manifestos covered in *Manifesto Corpus Translations* corpus (Ivanusch and Regel 2024), we demonstrate in the Supplementary Material (Section D.1) that the attribute emphasis profiles of Social Democratic and Conservative parties observed in these four countries closely resemble those of the broader population of Social Democratic and Conservative parties in a wider range of countries.

Taken together, our complementary sampling strategies maximize the diversity of observed social group references while balancing the substantial annotation effort required to develop a complex measurement framework. In total, we processed 494 party manifestos (202 Green, 182 PRR, 56 Social Democratic, and 54 Conservative) into a corpus comprising 436,941 sentences. Figure A1 and Figure A2 show an overview of the number of manifestos and sentences by party family included in our corpus.

4.2 Measurement

We proceeded in two steps to quantify the attributes used in social group mentions in this corpus. We first identified social group mentions in manifesto sentences, applying the methods described in Licht and Szczepanski (2025). We then labeled the extracted social group mentions by classifying the attributes they contain with a custom multi-label classification approach. In each step, we collected original manual annotations for a sample of texts to produce labeled data for (few-shot) supervised machine learning. While we build on established methodology in the first step, the classifiers developed for the second step are novel.

Social group mention detection

To be able to study empirically which types of attributes political parties emphasize when they mention social groups in their manifestos we first needed to identify and extract these social group mentions from manifestos’ texts. The task of social group mention detection involves locating and extracting any verbatim words or phrases from a text that refer to or

describe a social group and can be reliably automated through supervised token classification (Licht and Sczepanski 2025; cf. Horne et al. 2025). We thus base our approach for this measurement task on the supervised token classification approach.

However, our manifesto sentences corpus covers a much wider range of countries and periods than covered by existing labeled data or the training data of pre-trained classifiers.⁷ We thus collected original manual group mention annotations for our application following the approach proposed by Licht and Sczepanski (2025), which combines sequence annotation (i.e., marking relevant phrases in sentences) and supervised token classification to extract any verbatim words or phrases used in a sentence to refer to social groups. The primary challenge of working with text data from 36 countries was the broad language coverage of the corpus at hand, creating a bottleneck regarding the language skills of the human annotators we could recruit. We therefore machine-translated all non-English sentences in our manifesto corpus into English using the open-weights M2M model (Fan et al. 2020; Licht et al. 2026; cf. Licht and Lind 2023).⁸ Licht et al. (2026) find that applying transformer fine-tuning with machine-translated texts typically results in equivalent classifier performance and example-level predictions as training on original (multilingual) texts. The language coverage of this corpus is reported in Figure A4.

We proceeded in three annotation rounds (detailed in Section B), building on the ideas of active learning (Miller et al. 2020) and diversity sampling (Kaufman 2024) with the goal of maximizing the reliability and generalization of supervised classifiers trained on the collected annotations.⁹ Importantly, we stratified by manifesto in every annotation round to ensure broad coverage of countries, parties, and source languages. Thus, while we removed the language barrier for annotation through machine translation, we still accounted for the diversity of our target corpus when creating our annotated data.¹⁰

⁷Licht and Sczepanski (2025) focus their application on British party manifestos and complement it with exploratory applications in British parliamentary speeches and German party manifestos. Horne et al. (2025) annotated samples of English (GBR and IRL) and German (DEU and AUT) party manifesto sentences and validated their fine-tuned models additionally in Danish, Dutch, and Spanish samples.

⁸We created sentence-level data from raw manifesto texts using the `stanza` Python library.

⁹To mirror the approach by Licht and Sczepanski (2025), we distributed manifesto sentences’ English translations for annotation to allow fine-tuning a monolingual base model.

¹⁰Figure B1 and Figure B2 report the time and country coverage of the sentence we annotated for group mentions.

Combined, we collected approximately 7.8 thousand annotated sentences across our three annotation rounds. We already stopped after the third annotation round – having annotated only 0.28% of all sentences in the target corpus – because we observed that the span-level F1 score for detecting social groups converged to the estimates reported by Licht and Szczepanski (2025) and because their “learning curve” analysis suggested that adding more labeled training examples would have yielded diminishing returns. More generally, in transfer learning with transformer models, the absolute number and quality of annotated examples matter more than their proportion of the target corpus. Accordingly, we prioritized quality and diversity over quantity through our step-wise sampling strategy combining informativeness- and classification uncertainty-based sampling (cf. Licht et al. 2025).

To train the final group mention detection classifier used in our application, we combined all sentences annotated in these three rounds, set aside 15% of sentences for evaluation, and fine-tuned a RoBERTa base model for token classification.¹¹ The final classifier trained on our data achieved a span-level F1¹² of 0.927 and a sentence-level F1 of 0.981 in held-out sentences, which is on par with the performance reported by Licht and Szczepanski (2025) despite our broader and more diverse case selection.

Applying the resulting classifier to all sentences in our corpus yielded 209,351 predicted social group mentions. These are 68,348 unique mentions (i.e., unique verbatim phrases referring to social groups) when ignoring mentions’ sentence context. The six most common mentions are “people” (N=6,001), “children” (N=4,262), “women” (N=3,941), “citizens” (N=3,169), “society” (N=3,056), and “young people” (N=2,504).

Figure 1 summarizes the distributions of manifesto-level group mention prevalence estimates for the four party families covered by our corpus. For each party manifesto, we compute the prevalence of social group mentions as the share of its sentences that contain at least one social group mention according to our classifier. The box plots summarize the distribution of these manifesto-level estimates by party family. This shows that typically between 30 and 40% of sentences contain one or more social group mentions in the sample of

¹¹The models are publically available for download at huggingface.co/collections/haukelicht/social-group-mention-attribute-classification.

¹²The span-level F1 is 0.831 when only considering exact span matches (as per the strict `seqeval` metric, cf. Licht and Szczepanski 2025).

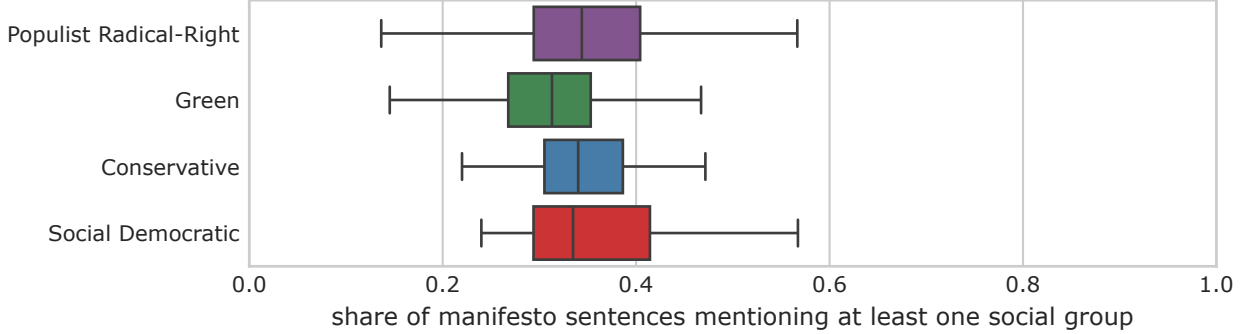


Figure 1: Share of manifesto sentences mentioning at least one social group, by party family based on social group mentions extracted with our fine-tuned RoBERTa model. The boxplots show the distribution across manifestos within each party family, excluding outliers.

manifestos we analyze.

Attribute classification

To classify the attributes contained in the social group mentions identified by our supervised mention detection classifier in the previous step, we applied our taxonomy to collect multi-label annotations for a sample of mentions.

Annotation Our coding instrument shows one mention at a time in its sentence context.¹³ Its coding logic reflects the structure of our taxonomy (cf. Table 1), which organizes attribute dimensions and categories across levels of abstraction and allows classifying a mention into multiple attribute categories if the group it refers to or describes is characterized by more than one kind of attribute. Accordingly, we first ask the coder whether the group mention qualifies as a “universal” group reference, and, if not, prompt them to select all economic and non-economic attribute categories that are contained in the group mention. A coder’s classification of a mention as “universal” rules out classification of the mention’s economic and non-economic attributes. However, any non-universal mention could be labeled with one or more of the available attribute categories.

Two trained research assistants independently annotated the attributes contained in mentions in our sample. Based on prior work (e.g., Thau 2019; Huber 2021; Dolinsky et

¹³See Section C.1 for details.

al. 2025), we expected some attribute categories to be much less common. We therefore used several annotation rounds to address label class imbalance and diversify the annotated sample.¹⁴ We detail these steps in the Supporting Materials, Section C, and summarize the main points here. The first rounds focused on sampling diverse and difficult cases and balancing attribute prevalence. Inter-annotator agreement was generally high, except in rounds where we strategically oversampled difficult cases or rare categories (see Table C1 and Table C2). To further improve annotation quality, the author team arbitrated disagreements, conducted a consolidation round, and reviewed cases where classifier ensembles disagreed with annotations.

In total, we collected, reviewed, and consolidated multi-label attribute annotations for 600 mention-in-context examples. As we demonstrate below, few-shot learning enabled us to train reliable classifiers with these data. The comparatively small sample reflects the resource-intensive nature of developing and implementing our novel coding scheme. Developing an adequate coding instrument and guidelines required substantial upfront investment in testing and coder training. Moreover, annotating 600 examples amounts to an estimated 6,900 coding decisions per annotator, since each non-universal mention must be evaluated against all 14 attribute categories.¹⁵ However, we expect that the labeled data and fine-tuned classifiers we contribute will enable future work to scale the human annotation process we propose more efficiently.

Given the importance of consistently labeled examples for few-shot learning, we prioritized annotation quality over quantity. The authors therefore arbitrated disagreements, conducted a consolidation round, and reviewed cases where classifier ensembles disagreed with annotations (see Section C). The resulting labeled data and fine-tuned classifiers should enable future research to scale this annotation process even more efficiently.

Automation The annotated mention-in-context examples covered only a small fraction of the 68,341 unique predicted social group mentions. We thus needed to automate attribute

¹⁴Label class imbalance is common in social science classification and can harm predictive performance. Iterative sampling guided by prediction uncertainty can help mitigate this.

¹⁵At an estimated prevalence of universal mentions at 25%, 150 of 600 examples only require one coding decision (universal or not) whereas the other 450 examples required the universal-or-not decision *plus* one binary decision (Yes/No) for each of our 14 attribute categories ($= 450 \times (1 + 4 + 10) = 6,750$).

classification for all mentions.

Current computational text analysis methods offer two main approaches for multi-label classification with few annotated examples: few-shot fine-tuning with encoder-only models (Devlin et al. 2019) or few-shot in-context learning with decoder-only large language models (Brown et al. 2020). We chose the first for its efficiency and proven effectiveness (cf. Tunstall et al. 2022), leaving the second for future work.

Specifically, we relied on the few-shot sentence transformer fine-tuning (SetFit) approach¹⁶ proposed by Tunstall et al. (2022) to train two separate multi-label classifiers – one for economic attributes and one for non-economic attributes. In the SetFit framework, labeled examples in the training set are first used to construct pairs for contrastive embedding model fine-tuning. The contrastive embedding finetuning task practically augments our 600 annotated examples to several thousand contrastive pairs, providing ample training material for embedding model fine-tuning that effectively tunes the text representations that go into the next step: training an attribute category multi-label classification model. The classification head is the component that maps a mention’s embedding into a multi-label attribute classification. Both components are trained end-to-end for multi-label classification on labeled examples.

Having thoroughly evaluated several implementation choices, we found that inputting only the group mentions’ text without the sentence context and using `all-mpnet-base-v2` as the base embedding model (Reimers and Gurevych 2019) performed best in our application. Details about our implementation can be found in Section C.3.

Table 2 summarizes the performance of the resulting attribute category classifiers averaged across five held-out test folds.¹⁷ The economic attribute classifier achieves a macro (micro) average F1 of 0.8 (0.785); the non-economic attribute classifier a macro (micro) average F1 of 0.804 (0.829). We consider the performance of our classifiers to be strong given the low number

¹⁶Tunstall et al. (2022) show that the SetFit approach enables effective few-shot learning in text classification applications. SetFit is also more compute-efficient at prediction time compared to natural language inference (NLI) used by Horne et al. (2025) (Laurer et al. (2024)), which multiplies the number of inferences needed by the number of label categories. Relying on SetFit thus allows faster and more resource-saving automated coding of unlabeled social group mentions.

¹⁷These are the results of models trained on folds’ respective training examples using “optimal” hyperparameters identified using hyperparameter grid search and early stopping based on macro-averaged F1 on the fold’s validation examples.

Table 2: Evaluation results of group mention attribute classifiers. Values report the mean precision, recall, and F1-score averaged across five folds for each attribute category, along with the prevalence of each category in the training set.

	F1-Score	Precision	Recall	prevalence
economic attributes				
micro average	0.785	0.822	0.758	
weighted average	0.784	0.831	0.758	
macro average	0.800	0.842	0.788	
<i>education</i>	0.942	0.933	0.967	0.033
<i>employment status</i>	0.721	0.770	0.733	0.059
<i>income/wealth/economic status</i>	0.751	0.857	0.675	0.093
<i>occupation/profession</i>	0.787	0.809	0.776	0.166
non-economic attributes				
micro average	0.829	0.881	0.782	
weighted average	0.826	0.896	0.782	
macro average	0.804	0.871	0.768	
<i>age</i>	0.872	0.918	0.841	0.106
<i>crime</i>	0.764	0.842	0.719	0.052
<i>ethnicity</i>	0.921	1.000	0.860	0.065
<i>family</i>	0.905	0.873	0.972	0.098
<i>gender/sexuality</i>	0.765	0.825	0.725	0.074
<i>health</i>	0.732	0.863	0.666	0.056
<i>nationality</i>	0.819	0.890	0.764	0.128
<i>place/location</i>	0.750	0.837	0.702	0.033
<i>religion</i>	0.759	0.770	0.776	0.050
<i>shared values/mentalities</i>	0.753	0.889	0.660	0.135

of few-shot training examples, strong label class imbalance, and novelty of our taxonomy. The classifiers prove similarly reliable as trained human annotators (see Table C2) in classifying economic attributes and even more reliable in classifying non-economic attributes. And while classification reliability varies across attribute categories, it is overall very good, with most categories achieving an F1 of 0.75 or higher. Moreover, the variation in classification reliability across attribute categories is not correlated with attribute categories’ prevalence in the training data (see last column of Table 2), which suggests that performance differences are explained by task difficulty rather than mere lack of training material. Nevertheless, we expect that future work will be able to improve on our results by collecting more (informative) training data, for example by leveraging our pre-trained classifiers or LLMs as (collaborative)

annotators. And measurement error introduced by our classifier in downstream regression analysis of specific attribute categories can be adjusted using error correction frameworks like design-based supervised learning (Egami et al. 2023).

Validation To assess the validity of our classifiers, we conduct three analyses reported in full in Section C.4. First, a “fighting words” lexical analysis (Monroe et al. 2008) shows that the most distinctive terms for each attribute category align strongly with substantive expectations. For instance, *education* is characterized by terms like “students”, “pupils”, “school”, and “university”, while *shared values/mentalities* features value-laden adjectives and lemmas of “society”. However, despite the distinctiveness of these terms, our analysis shows that simple keyword dictionaries cannot reliably replace our classifiers: the recall of the top-10 most distinctive terms is below 50% for attribute categories such as *religion* (42%), *nationality* (36%), *income/wealth/economic status* (33%), and *occupation/profession* (20%). These findings underscore the validity of supervised classification in our application.

Second, comparisons to group reference classifications in existing categorization schemes (Thau 2019; Horne et al. 2025) underscore the added value of our attribute-centered, multi-label, hierarchical taxonomy. Comparing our attribute-centered group reference classifications to those in Thau (2019) reveals strong convergence where expected (e.g., 91% of his *Age/generation* mentions are classified as *age* by our classifier, and 87% of his *Professional group* mentions as *occupation/profession*), but also that his single-label scheme conceals meaningful attribute combinations: mentions in a given Thau category are frequently assigned to multiple attribute categories in our scheme. Comparing our classifications to those obtained with the multi-label group category scheme of Horne et al. (2025), in turn, shows that both schemes partition the social group mentions in our corpus overall very similarly, as indicated by expected positive associations between occupation-related categories and systematic negative associations between non-economic and occupation/class-based categories. However, our attribute-centered framework decomposes some of Horne et al.’s broad categories (e.g., *Ethnic And National Communities*) into finer-grained attribute combinations while grouping very specific ones (e.g., their many specific occupational group categories) while also capturing group mentions not covered by any of their pre-defined categories. Taken together,

these analyses provide strong evidence of the content validity of our classifiers and clarify the analytical added value of our attribute-centered, multi-label measurement approach.

5 Prevalence of attribute dimensions, categories, and their combinations

We now turn to descriptive statistics of the attribute classifications we have obtained for all social group mentions extracted by our group mention detection model in the sentences of manifestos in our corpus. Note that we include in this analysis also group mentions in manifestos of centre-left and -right parties in the four countries covered by our case selection.

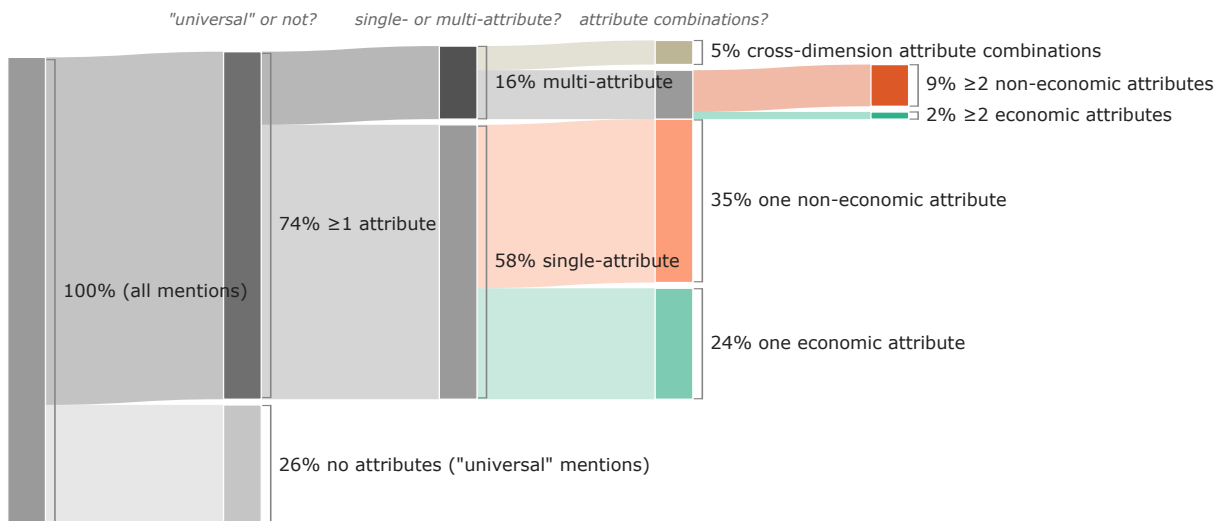


Figure 2: Sankey diagram showing the breakdown of social group mentions by attribute prevalence and attribute combinations. The leftmost column shows the total number of mentions extracted in our manifesto sentences corpus, which are then split into "universal" mentions (no predicted attributes) and mentions with at least one predicted attribute. The latter are further divided into single-attribute and multi-attribute mentions, with multi-attribute mentions split into those with only within-dimension combinations and those with cross-dimension combinations. The rightmost column shows the breakdown of within-dimension combinations into economic and non-economic attributes.

First, Figure 2 presents a breakdown of the distribution of the number of attribute categories into which our classifiers assigned the social group mentions we have identified in

our corpus.¹⁸ This shows that the majority of social group mentions (74%) we identify in the manifestos in our corpus feature at least one attribute covered by our taxonomy. The rest are “universal” mentions according to our definition, meaning that they do not feature any economic or non-economic attributes in our taxonomy.

Among the 74% non-universal social group mentions, slightly less than four out of five mentions (79%) are predicted to feature only a single attribute (58% of the total population). About 60% of these mentions feature a non-economic attribute and 40% an economic attribute – a first indication that non-economic attributes are more frequently invoked to refer to social groups than economic attributes. However, 21% of social group mentions we classify as non-universal (16% of all mentions) are predicted to feature more than one social attribute.

We turn next to the overall prevalence of individual attribute categories among non-universal mentions. The prevalence of an attribute category is the share of social group mentions predicted to feature that category.

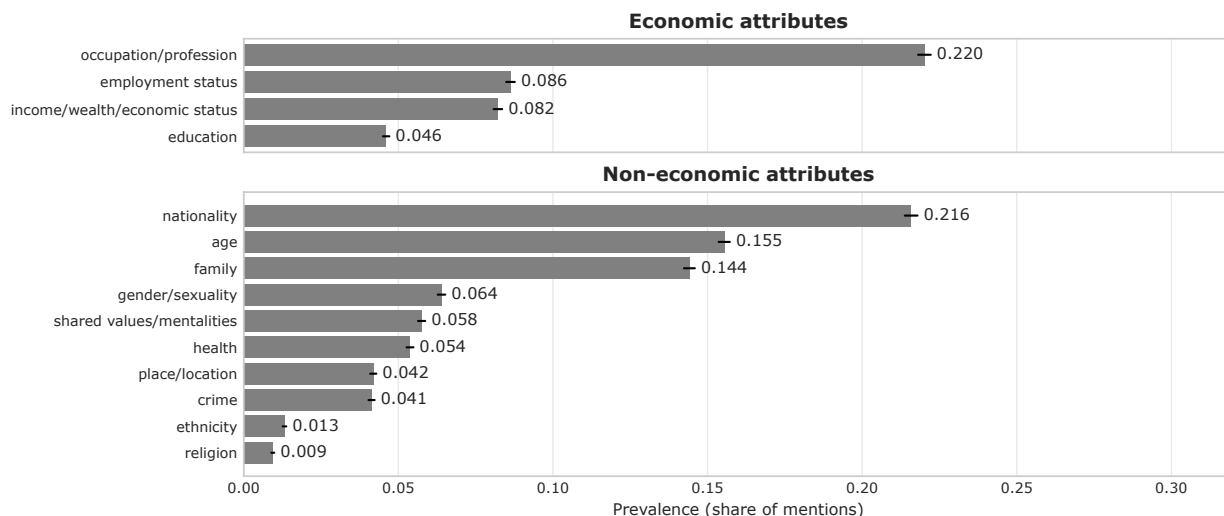


Figure 3: Prevalence of attribute categories across all social group mentions (excluding universal mentions with no specific attributes). Bars show the share of mentions featuring each attribute, with 95% confidence intervals. Top panel: economic attributes; bottom panel: non-economic attributes.

Figure 3 shows that *occupation/profession* is the most prevalent economic category and *nationality* the most prevalent non-economic category. The second and third most prevalent economic attributes (*employment status* and *income/wealth/economic status*) are relatively

¹⁸We use a 0.5 threshold for predicted probabilities of attribute categories.

far less prevalent than the second and third most prevalent non-economic attributes (*age* and *family*), which show substantially higher prevalence.

Notably, the non-economic *shared values/mentalities* category we introduce is predicted to occur in almost 6% of non-universal social group mentions, and thus more often than the least prevalent economic attribute category (*education*) as well as more often than five out of the remaining nine non-economic attribute categories. This underscores the added value of introducing this novel category in our taxonomy. We discuss attribute category prevalence again when focusing on differences across party families in Section 6.

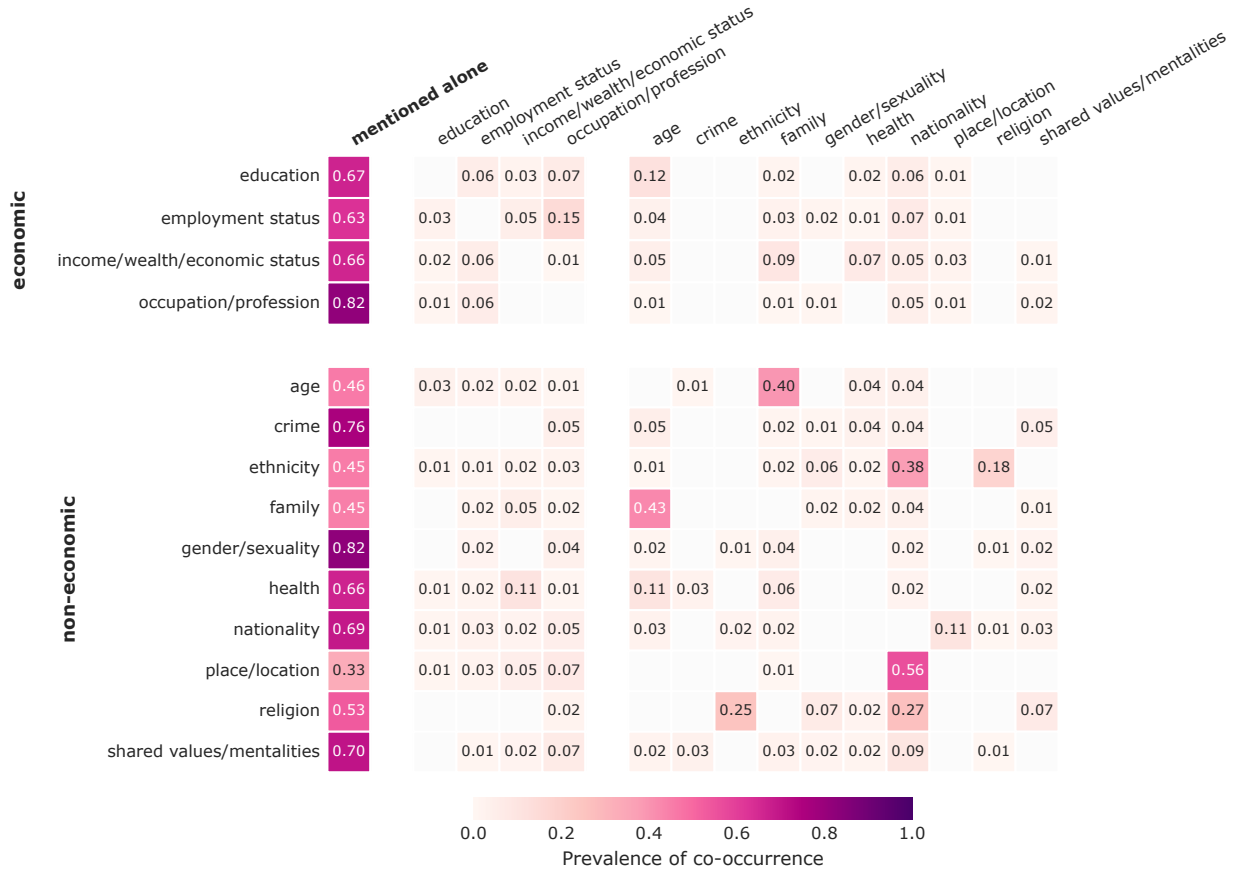


Figure 4: Co-occurrence patterns of attribute categories in social group mentions. Heatmap cells show the share of mentions where each focal attribute (rows) co-occurs with other attributes (columns). The first column ('mentioned alone') shows the share of mentions where the focal attribute appears without any other attributes. Top panel: economic attributes; bottom panel: non-economic attributes. Values below 0.01 are not displayed.

To gain a more granular understanding of attribute combination patterns at the group mention level in our data, we next compare attribute categories' co-occurrences. Recall that

Figure 2 showed that if a party uses attributes to invoke social groups in their manifestos, it combines attributes from different categories about one out of five times. Among such multi-attribute mentions, about two thirds feature “within-dimension” attribute combinations, that is, combinations of attributes from categories of the same attribute dimension (e.g., combinations of *education* and *employment status* as in “students, who are working longer hours to fund their university study”). Such within-dimension attribute combinations are about four times more frequent among non-economic attributes, which – viewed together with the higher prevalence of non-economic attributes – underscores the importance of non-economic attributes in parties’ social group mentions. In contrast, we estimate that every third multi-attribute social group mention combines economic with non-economic attributes (e.g., *employment status* and *health* as in “persons with disabilities employed in a sheltered workshop”).

The first plot panel column in Figure 4 reports how likely each attribute category appears alone in a group mention. This reveals substantial variation across mentions in different attribute categories in their tendencies to be single-attribute. Among economic attributes, *occupation/profession* is mentioned most frequently as the only kind of attribute in a social group mention (82%). This is about 1.3 times more often than the economic attribute that is most frequently combined with attributes from other categories: *employment status*, which occurs alone in only 63% of mentions that feature attributes of this category. Among non-economic attributes, *gender/sexuality* is mentioned most frequently alone (82%) – almost 2.5 times more often than *place/location* (33%).

Focusing on the two right-most heatmap columns in Figure 4, which show attribute categories’ co-occurrence tendencies, moreover allows examining specific combination patterns.¹⁹ This view reveals that most economic within-dimension attribute combination is driven by the combination of *employment status* with other economic attributes. For non-economic attributes, the stronger tendency of within-dimension attribute combination is driven by the frequent co-occurrence of *age* and *family*, and of *ethnicity*, *place/location*, or *religion* with *nationality*.

¹⁹Figure D4 reports an alternative visual presentation of co-occurrence patterns by analyzing co-occurrence patterns using *conditional probabilities* of co-occurrence for all attribute category pairs. Table D1 shows concrete examples sampled from our corpus.

Regarding cross-dimension attribute combinations, economic attributes are frequently combined with *age* (especially *education*), *family*, *health*, and *nationality*. The non-economic attribute *health* is often combined with *income/wealth/economic status*. And *place/location* and *shared values/mentalities* are often combined with *occupation/profession*.

These patterns are particularly interesting because they add depth to the analysis of parties' group attribute focus. For example, while social group mentions featuring descriptions of *shared values/mentalities* are comparatively rare, their tendency to co-occur with *occupation/profession* and *nationality* suggests that parties use such attributes to differentiate their group focus strategies and specify which kind of people they mean exactly when referring to groups.

6 Understanding social group references across analytical levels

The taxonomy developed in this paper not only provides a framework for classifying social group references but also clarifies the analytical level at which different theories of party-group relations make their primary claims. Rather than viewing these perspectives as competing explanations, we argue, in line with Westheuser and Zollinger (2025), that they primarily emphasize different – yet partly overlapping – layers in the political formation of social groups. Distinguishing these layers allows their theoretical expectations to be evaluated separately while avoiding the incorporation of theoretical assumptions directly into the measurement of social groups.

Socio-structural approaches to party competition – including cleavage theory, dealignment, and more recent accounts of partisan realignment – are primarily concerned with which social groups become politically salient, expecting parties to mobilize groups that reflect enduring social structures and evolving electoral coalitions (Lipset and Rokkan 1967; Dalton 1984, 2008; Kriesi et al. 2008, 2012; Piketty 2018; Gethin et al. 2021). Constructivist approaches, in turn, shift attention from the structural existence of social groups to the political processes through which they are constituted and acquire political meaning (Chandra 2012; de Leon

2014; Wolkenstein and Wratil 2021; Zuber et al. 2025). Because the selection, combination, and interpretation of social attributes can themselves shape how political conflict is framed, constructivist approaches naturally overlap with theories of political framing and identity formation (Westheuser 2025; Westheuser and Zollinger 2025; Zollinger 2022, 2024).

A related body of work focuses primarily on how already constituted social groups are politically interpreted and evaluated. Research on deservingness and the social construction of target populations examines whether groups are portrayed as deserving or undeserving, threatening or vulnerable, productive or dependent, or as members of a shared political community (Schneider and Ingram 1993; Oorschot 2000, 2006). These approaches generally treat the referenced groups as analytically given, focusing on their evaluation rather than on the processes through which they are defined (Theiss 2023; Deiss-Helbig et al. 2026; Esmark and Schoop 2017). Political framing, however, is broader than this final evaluative layer. Parties can frame social conflict by selecting which group-defining attributes to invoke, by combining attributes into particular group constructions, and by attaching evaluative meanings to the resulting groups (cf. Entman (1993)). Our analytical distinction therefore does not imply that framing occurs only at the evaluative layer; rather, we distinguish the evaluation of already constituted groups from the selection and combination of attributes through which groups are defined.

By separating the attributes through which groups are defined from their subsequent evaluation, while recognizing that political framing can operate across both processes, our taxonomy provides a common measurement framework within which these perspectives can be examined individually or in combination. In the following, we illustrate this analytical distinction by deriving expectations from each perspective and comparing them to the patterns observed in party manifestos.

Socio-structural expectations

Socio-structural theories imply that differences in party competition should primarily become visible in the selection of social attributes. Because parties mobilize electoral coalitions rooted in different social positions and conflict structures, they are expected to differ systematically in the types of groups they invoke (Lipset and Rokkan 1967; Kriesi et al. 2008, 2012; Bornschieer

2015; Piketty 2018; Gethin et al. 2021). The attribute-centered nature of our taxonomy allows these expectations to be examined across multiple levels of abstraction, from broad attribute dimensions to increasingly specific group references.

For example, Green parties are expected to place greater emphasis on social groups associated with the winners of post-industrial transformation, including highly educated professionals, younger cohorts, women, and groups linked to emerging socio-cultural conflicts (Marks et al. 2018; Bornschier 2015; Westheuser and Zollinger 2025). Populist radical right (PRR) parties, by contrast, should emphasize nationally bounded groups, manual and technical occupations, and groups associated with the economic and cultural consequences of globalization, educational expansion, and immigration (Mudde 2007; Kriesi et al. 2008; Stubager 2009).

These expectations should become increasingly pronounced across the hierarchy of our taxonomy. At the level of attribute dimensions, both Green and PRR parties should rely more strongly on non-economic attributes than their centre-left and centre-right competitors, reflecting the growing importance of socio-cultural conflicts in contemporary party competition. At the level of attribute categories, however, clearer differences should emerge. Green parties are expected to place greater emphasis on gender and sexuality, age, and socio-cultural occupations, whereas PRR parties should focus more strongly on nationality and occupations associated with production, security, and traditional labour markets. Finally, at the level of specific group references, these differences should translate into distinct repertoires of groups, such as women, LGBTQI+ persons, students, and socio-cultural professionals for Green parties, compared with manual workers, farmers, police, entrepreneurs, or nationally defined communities for PRR parties.

Figure 5 compares the prevalence of attribute dimensions and categories across Green, PRR, centre-left, and centre-right parties. Contrary to a common interpretation of contemporary party competition, the four party families differ little in the overall balance between economic and non-economic group references. Across all party families, non-economic attributes occur roughly one and a half to two times more frequently than economic attributes, while references lacking either type of attribute (universal references) occur at comparable rates. Thus, the prominence of non-economic group references is not distinctive to Green

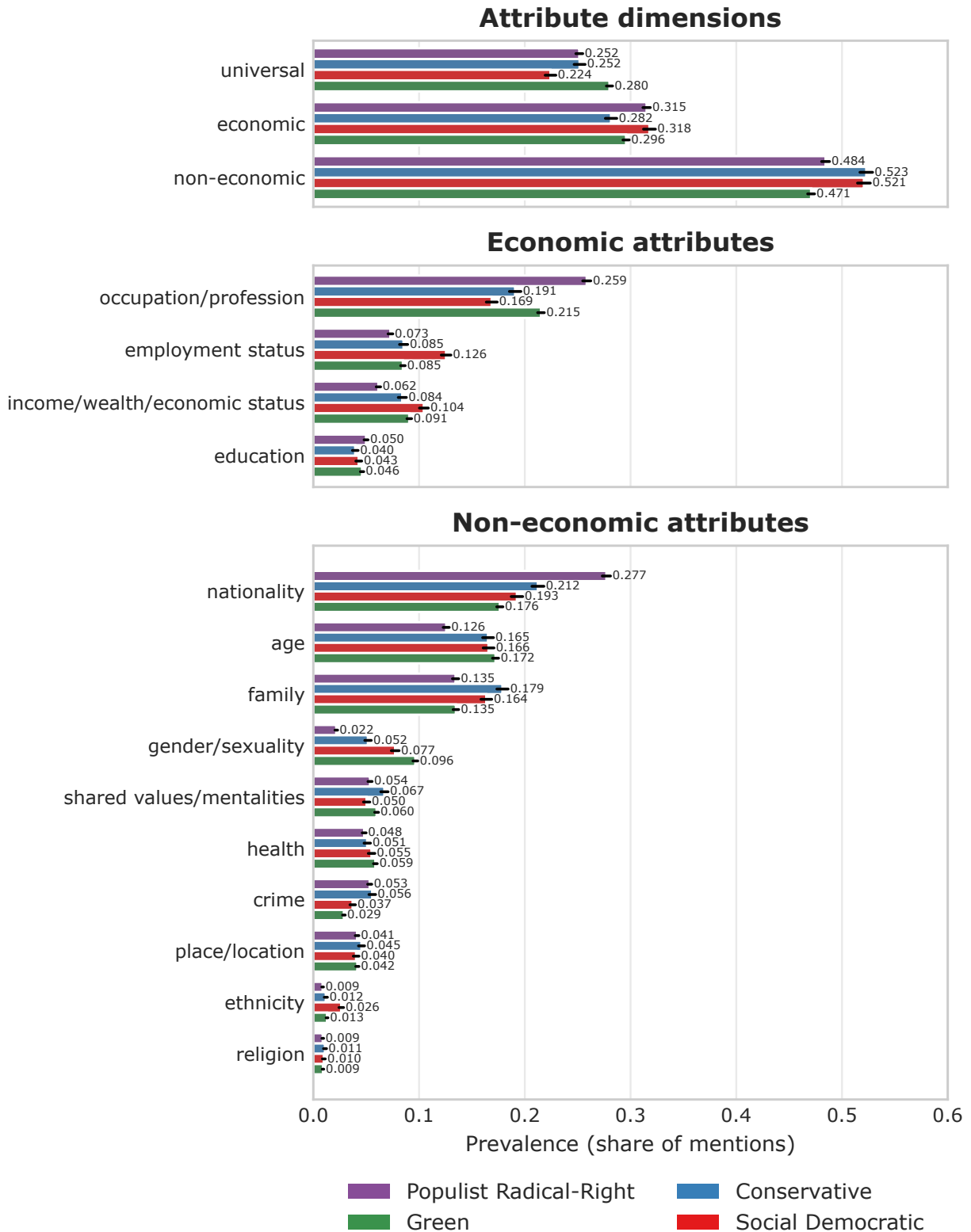


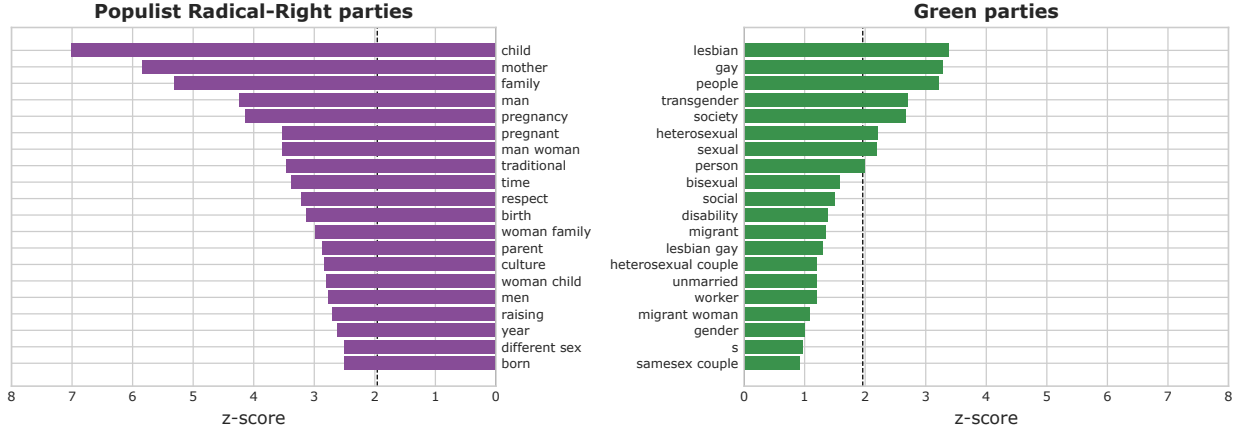
Figure 5: Prevalence of attribute dimensions in all mentions and economic and non-economic attributes in non-universal social group mentions in parties' election manifestos by party family. Bars show share of mentions containing each attribute, with 95% confidence intervals. *Note:* Top panel shows prevalence in all mentions, while middle and bottom panels show prevalence in mentions containing at least one attribute.

and PRR parties but characterizes centre-left and -right parties as well.

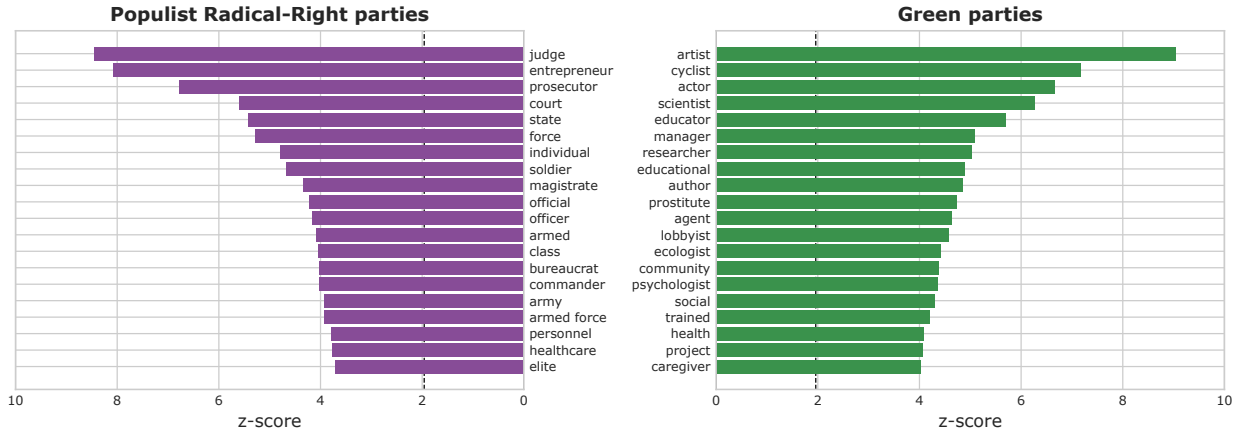
Differences emerge, however, once attention shifts from broad attribute dimensions to specific attribute categories. Within the economic dimension, both Green and PRR parties rely more heavily on occupational attributes than their mainstream competitors, although this tendency is considerably stronger among PRR parties. By contrast, PRR parties place comparatively little emphasis on employment status and income-related attributes, distinguishing them particularly from social democratic parties. Within the non-economic dimension, Green parties attach substantially greater importance to gender- and age-related attributes, whereas PRR parties are distinguished by their strong emphasis on nationality. Interestingly, references to family-related groups are most common among centre-left and -right parties rather than among the challenger parties.

These patterns remain remarkably stable over time (see Figure D1 and Figure D2) and closely resemble those observed in a sample covering a wider range of Conservative and Social Democratic parties (see Figure D3), suggesting that they are not an artefact of the party sample used for the main analysis.

Moving to the most specific level of the taxonomy further illustrates how party families differentiate through the concrete groups they invoke. To examine how parties populate a given attribute category with different social groups, we employ fighting words analysis (Monroe et al. 2008), which identifies the terms that are statistically most characteristic of one party family relative to another within a given subset of group references. The fighting words analyses in Figure 6 show that Green parties' gender- and sexuality-related group references are characterized by terms relating to homosexuality, non-binary gender identities, and intersections with health and migration. PRR parties, by contrast, associate gender primarily with fertility, birth, childrearing, and traditional family roles. Similar differences emerge for occupational references. PRR parties most distinctively invoke members of the armed forces, entrepreneurs, bureaucrats, and the judiciary (cf. Dolinsky et al. 2025), whereas Green parties emphasize socio-cultural and educational professionals, intellectuals, and lifestyle-related groups.



(a) Gender/sexuality-related social group mentions



(b) Occupation/profession-related social group mentions

Figure 6: Most distinctive words of group mentions in selected attribute categories in Populist Radical-Right and Green parties’ election manifestos. Values plotted are z -scores from “fighting words” analysis. Values above ± 1.96 (vertical dashed line) can be considered significantly distinctive.

Constructivist expectations

Constructivist approaches imply that differences between parties become particularly visible not in the prevalence of broad attribute categories but in the ways attributes are combined to construct politically meaningful social groups. Rather than treating social groups as fixed expressions of underlying social structures, this literature emphasizes that political actors actively constitute groups by selecting, combining, and symbolically interpreting social attributes (Chandra 2012; Brubaker 2004; de Leon 2014; Wolkenstein and Wratil 2021; Zuber et al. 2025).

A central mechanism in this process is moral boundary-making. Political actors distinguish us from them not only through socio-demographic attributes but also through shared values, lifestyles, and moral dispositions (Lamont 1992, 2009; Wimmer 2013; Westheuser 2025; Westheuser and Zollinger 2025). Importantly, morality can play two analytically distinct roles. Moral attributes may themselves define a social group – for example, the “hard-working” or “ordinary” people – or they may combine with existing attributes to create narrower groups, such as hard-working immigrants. By contrast, moral *evaluations* as in “immigrants are dangerous” or “welfare recipients are lazy” do not redefine group membership but attach evaluative meaning to already constituted groups. Following the analytical distinction developed in this paper, we therefore treat the former as instances of group construction, whereas the latter belong to the subsequent layer of political evaluation.

These considerations imply two expectations. First, constructivist approaches should become visible in attribute compositions, particularly where moral or value-related attributes combine with socio-demographic attributes to define new groups. Second, they should become visible in groups constituted primarily through shared values, mentalities, or moral dispositions, because these attributes define collectivities independently of conventional socio-demographic characteristics.

Figure 7 examines attribute compositions in Green and PRR manifestos. Several clear differences emerge. PRR parties frequently combine nationality with occupational, family, crime, and especially shared-value attributes, thereby constructing nationally bounded social groups differentiated by moral characteristics. Green parties, by contrast, more often combine age, gender, and economic attributes within individual group references.

Particularly striking is the role of shared values and mentalities. Although both party families invoke value-based attributes with similar overall frequency (see Figure 5), PRR parties combine these attributes much more frequently with other defining characteristics, especially nationality and occupation. Rather than merely mobilizing existing socio-demographic groups, they therefore construct politically meaningful groups through combinations of structural and moral attributes.

Constructivist scholarship has argued that the universalist pole of contemporary political conflict is often articulated less explicitly than its particularist counterpart (Westheuser and

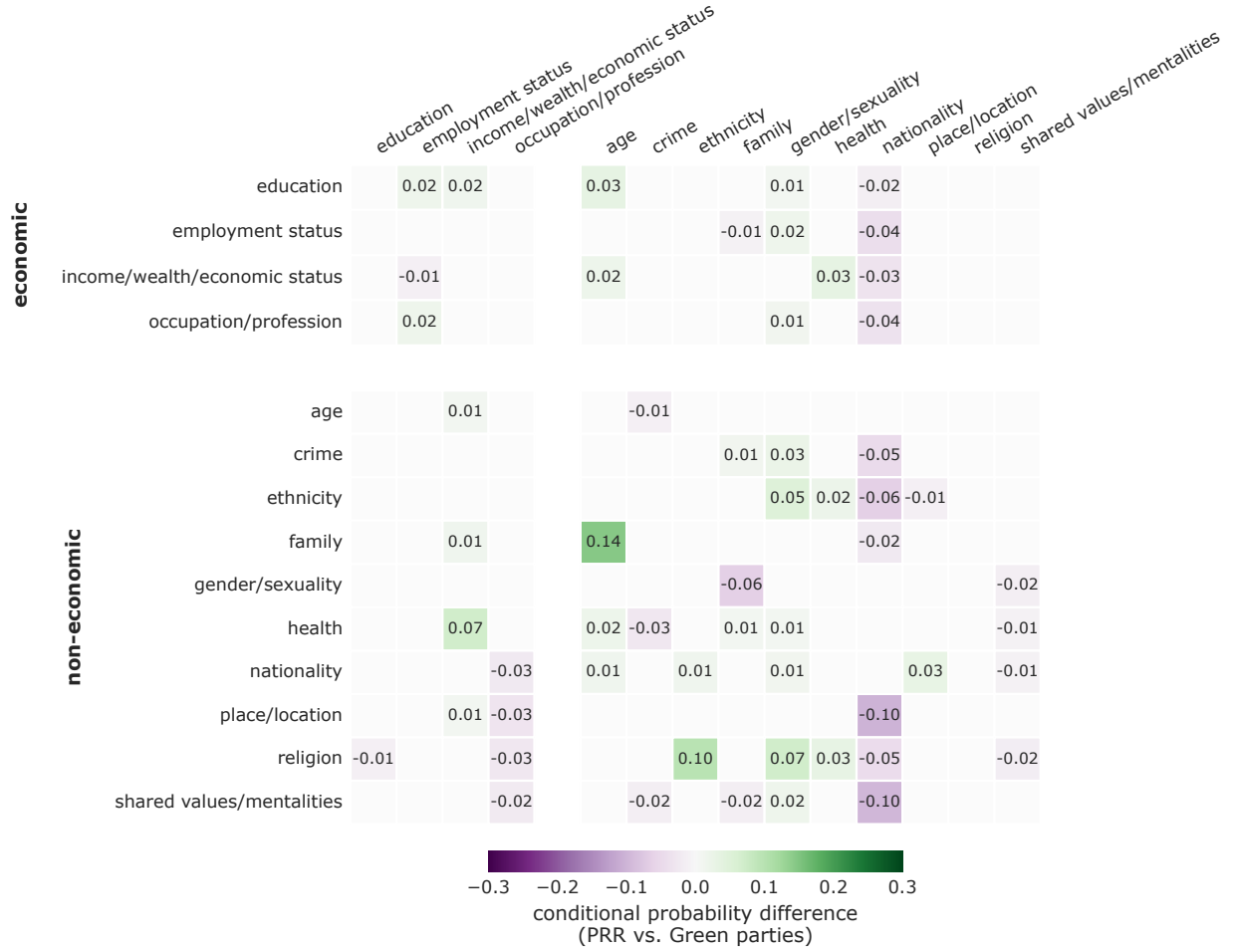


Figure 7: Differences in conditional probabilities of attribute co-occurrence between Green and Populist Radical-Right (PRR) party manifestos. Heatmap cells show the difference in $\Pr(B|A)$ between Green and PRR parties (Green – PRR). Rows indicate “attribute A” (the conditioning attribute) and columns indicate “attribute B” (the outcome attribute). Positive values (green) indicate that attribute B is more likely to be mentioned given attribute A in Green party manifestos compared to PRR manifestos, while negative values (purple) indicate the opposite. *Note:* Values below 0.01 in absolute value are not displayed.

Zollinger 2025). Combining our taxonomy with a fighting words analysis allows us to examine this question directly. Specifically, we focus on group references of Green and PRR parties in our corpus classified as involving shared values or mentalities that also invoke society.²⁰ This allows us to contrast the constitutive value attributes through which Green and PRR parties define this otherwise universal collectivity. Figure 8 shows that the two party families rely on

²⁰In particular, we focus on the group mentions in this subset that contain the lemmas of the terms “society” and “societies”.

markedly different repertoires of constitutive value attributes. Green parties predominantly define society through relational principles such as solidarity, sustainability, inclusion, justice, and openness, whereas PRR parties emphasize prosperity, freedom, health, activity, and national belonging. Although references defined by shared values account for only around five to seven percent of all group mentions, they occupy a particularly important position within constructivist theories because they reveal the competing moral grammars through which parties define society itself.

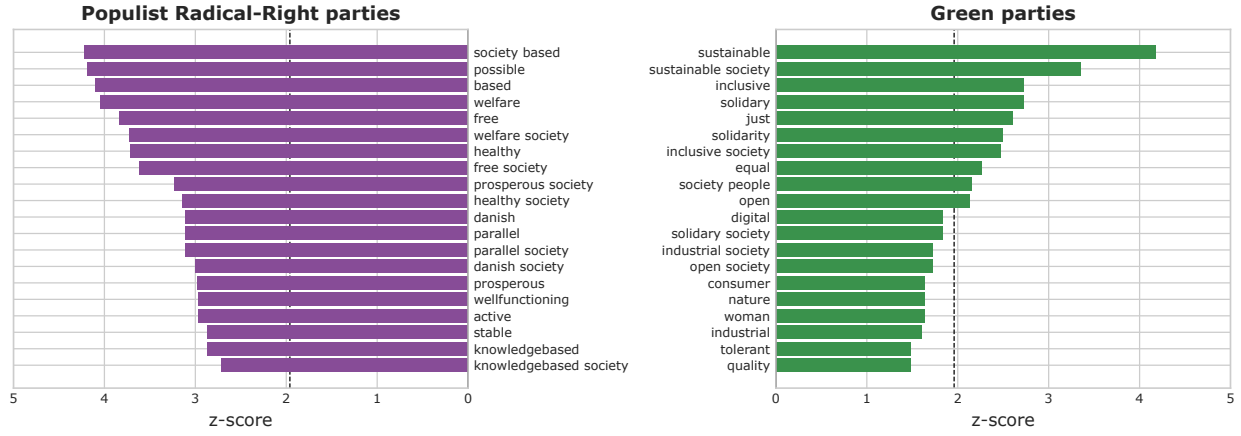


Figure 8: Most distinctive words for mentions of *shared values/mentalities*-based references to ‘society’ in Populist Radical-Right and Green parties’ election manifestos. Values plotted are z -scores from “fighting words” analysis. Values above ± 1.96 (vertical dashed line) can be considered significantly distinctive.

Group evaluation and deservingness

Approaches to deservingness and group evaluation shift attention from the definition and construction of social groups to how already identified groups are portrayed in political communication. Whether migrants are described as contributors or free riders, workers as deserving producers or disadvantaged victims, entrepreneurs as innovators or privileged elites, or LGBTQI+ people as valued members of society or objects of political contestation cannot be inferred from group-defining attributes alone (Entman 1993; Schneider and Ingram 1993; Oorschot 2000, 2006; Esmark and Schoop 2017).

Our taxonomy establishes the necessary analytical foundation for such analyses by systematically identifying the types of attributes emphasized in the referenced social groups.

Evaluating parties’ references to groups requires an additional analytical layer that captures the political stance or evaluative frame attached to each reference.

To illustrate this complementarity, we combine our attribute classifications with the stance classifier developed by Horne et al. (2025)²¹ and examine how Green and PRR parties evaluate group references featuring gender and sexuality attributes. Figure 9 first compares overall party stances towards these group references and subsequently distinguishes between the specific groups identified by Horne et al.’s taxonomy (men, women, and LGBTQI+ people).

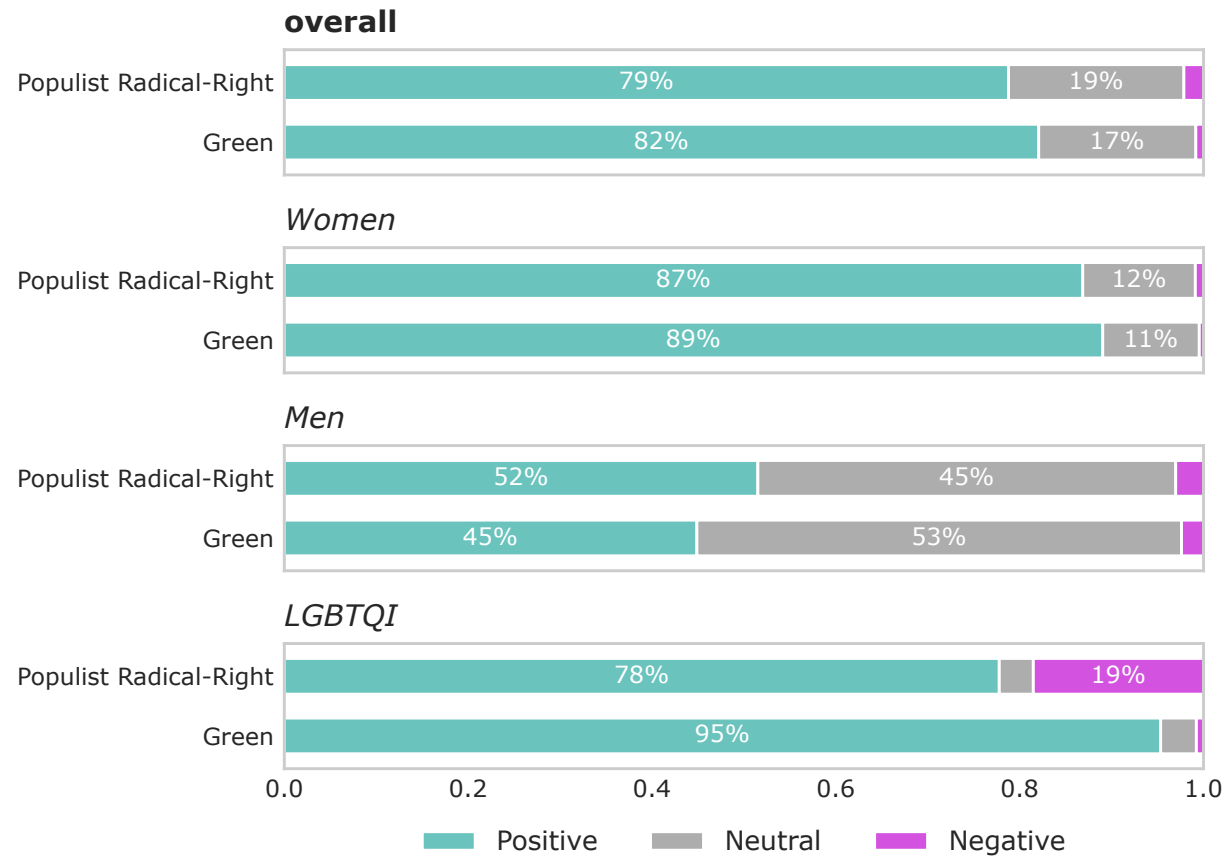


Figure 9: Stance distribution towards *gender/sexuality*-related groups by party family. Top panel shows overall distribution across all gender/sexuality groups. Subsequent panels show distributions for specific group categories (LGBTQI, Men, Women), with bars comparing stance distributions between Populist Radical-Right and Green parties.

At the aggregate level, Green and PRR parties differ surprisingly little. References to

²¹[rwillh11/mdeberta_NLI_stance_NoContext](#) (accessed on July 8, 2026).

gender- and sexuality-related groups are overwhelmingly positive in both party families, while explicitly negative references remain extremely rare. This finding echoes a broader pattern in the literature on party group appeals, which has repeatedly shown that parties typically invoke social groups in supportive or representative terms rather than to criticize them (e.g., Thau 2019; Huber 2021; Riethmüller and Franzmann 2025; Horne et al. 2025). This underscores that the prevalence of group references is an important dimension of group-based party competition – much like issue salience in studies of issue competition.

However, once attention shifts to specific groups, pronounced differences emerge. PRR parties express substantially more positive evaluations of men than Green parties, whereas Green parties overwhelmingly portray LGBTQI+ people positively. By contrast, roughly one in five PRR references to LGBTQI+ people is negative.

Together, these findings illustrate why our taxonomy’s separation of group identification from political evaluation is analytically productive: it allows socio-structural, constructivist, and deservingness perspectives to be examined within a common measurement framework.

7 Conclusion

This paper introduces an attribute-centered taxonomy for analyzing political parties’ references to social groups together with a scalable measurement framework for identifying these references in political text. By taking attributes rather than predefined social groups as the fundamental unit of classification, the taxonomy provides a common analytical language for studying how parties define social groups while remaining agnostic with respect to competing theories of party competition, representation, and group formation.

Beyond providing a new measurement framework, the taxonomy also clarifies the analytical layers at which different theories of party–group relations make their primary claims. Socio-structural approaches primarily explain which social attributes become politically salient. Constructivist approaches explain how parties combine and reinterpret these attributes to constitute politically meaningful groups. Approaches to deservingness and group evaluation focus on how these groups are subsequently portrayed and evaluated. More broadly, the political framing of social groups can operate across all three analytical layers: through the

selective invocation of group-defining attributes, their combination into particular group constructions, and the evaluative meanings attached to the resulting groups.

Our empirical application illustrates the analytical value of this distinction. Across more than five decades of party manifestos in 36 democracies, the taxonomy reveals that broad attribute dimensions conceal differences that become increasingly pronounced at more specific levels of abstraction. While Green, populist radical right, centre-left, and centre-right parties rely on remarkably similar overall proportions of economic and non-economic group references, they differ systematically in the specific attributes they prioritize and, even more strongly, in how they combine these attributes to construct politically meaningful groups. In particular, the analysis of attribute combinations and shared values demonstrates how parties actively organize familiar social attributes into competing conceptions of society that would remain difficult to detect using conventional group classifications.

Finally, the taxonomy establishes a common foundation for analyses that extend beyond the identification of social groups themselves. Combining attribute classifications with stance or group-evaluation analysis makes it possible to distinguish which groups parties invoke, how they constitute these groups, and how they evaluate them. More generally, the taxonomy provides a common organizational backbone for extending analyses to lower levels of abstraction. Deductively, occupational references can be linked to ISCO, educational references to ISCED, or multiple attribute categories combined into broader schemes such as Oesch's class typology. Inductively, approaches such as fighting words, topic modelling, or embedding-based clustering can uncover variation within attribute categories and identify emerging forms of group construction.

More broadly, our framework also speaks to a longstanding question in the study of party competition. Both classical cleavage research and recent constructivist and entrepreneurial accounts emphasize social groups and political identities as foundations of enduring ties between parties and citizens. What differs is the process through which these linkages are formed. Classical cleavage politics was sustained by dense organizational networks and socialization within relatively stable social milieus. In increasingly dealigned and realigned party systems, by contrast, these organizational foundations have weakened, making parties' mass communication a more important arena in which social groups are invoked, boundaries

are drawn, and political identities are constructed. Yet, despite this renewed attention to political social group identities, their formation remains difficult to observe empirically. As Westheuser and Zollinger (2025) put it, identity formation often resembles “ships passing in the night” (Westheuser and Zollinger 2025, 120). Political identities are unlikely to be directly observable in isolated text units. Instead, they emerge through the cumulative repetition of communicative practices: the recurring invocation of particular social groups, their constitution through combinations of attributes and symbolic boundaries, and their repeated evaluation through political framing. By separating these communicative building blocks, our taxonomy provides a common measurement framework for studying how repeated patterns of group construction and political evaluation crystallize into enduring party-group identities – and equally importantly, when they fail to do so.

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Disclosure of interest

The authors confirm that there are no relevant financial or non-financial competing interests to report.

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Data availability statement

The data and code supporting the findings of this study are openly available in Harvard Dataverse at DOI TO BE ADDED.

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